

Unit 04 Review (Forces & Newton's Laws)

$F_{NET} = ma$ $F_{NET} = \pm F_1 \pm F_2 \pm \dots$ $F_G = mg$ $g = 9.8 \text{ m/s}^2$

1. A penguin by the name of Pickles ($m = 23 \text{ kg}$) is standing on a scale in an elevator on the top floor of a building. The elevator begins to go down towards the ground floor and does so by accelerating downward at 2.7 m/s^2 . What is the normal force?

2. A new sports car ($m = 980 \text{ kg}$) is capable of applying a force of 5500 N with its engine. If the force of friction on the car is 500 N , what is the car's acceleration (assume no air resistance)?

3. You are pushing a large tub of chocolate, which has a mass of 455 kg , with an acceleration of 0.4 m/s^2 . The force of friction is 88 N . What is the force with which you are pushing the tub?

4. A fish ($m = 9.1 \text{ kg}$) is pulled straight up out of the water by a fisherman. What is the acceleration of the fish if the force applied by the fisherman is 120 N ?

Answers: (1) 163.3 N ; (2) 5.10 m/s^2 ; (3) 270 N ; (4) 3.39 m/s^2

Name:

Date:

Period:

5. Definitions:

(a) mechanical equilibrium

(b) inertia

(c) weight

(d) mass

(e) F_G

(f) g

6. List each of Newton's Laws

7. If Newton's third law is correct, then a tiny smart car feels the same force as a huge semi-truck when the two have a head on collision. Why would you rather be riding in the semi-truck?

8. Explain which of Newton's 3 laws best describes the situation.

(a) You are drinking a cup of coffee while driving. When you slam on the brakes, your coffee spills out.

(b) If you are standing in a boat, and step out onto the dock, the boat moves away from you.

(c) As you add more items to a grocery cart, you notice its acceleration decreases.

(d) You are wearing roller blades and stand facing a brick wall. You push forward, against the wall, but end up traveling backwards. Why?

9. What are the two types of friction and which one is stronger?

10. What is friction dependent on?